ABHISHEK RAGHUWANSHI

AI/ML Engineer | LLM & RAG Systems | Computer Vision | MLOps | IIT Bombay M.Tech
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PROFESSIONAL SUMMARY

IIT Bombay M.Tech AI/ML Engineer with production experience building LLM, RAG, and Computer Vision systems for the World Food Programme (WFP) and Ministry of Agriculture & Farmers Welfare. Delivered end-to-end AI pipelines serving 10,000+ users across AWS and GCP - from data engineering and model development to Docker-based cloud deployment.

EDUCATION

Indian Institute of Technology, Bombay *Jul'22 – Jul'24*
Master of Technology & Development | **CGPA: 8.09/10**

WORK EXPERIENCE

IRCC, IIT Bombay | Research Associate (AI/ML) *Feb'25 – Present*
Tech: Python, LangChain, LangGraph, GPT-4, FAISS, Vector DBs, HuggingFace, PyTorch, EfficientNet, OpenCV, Docker, GCP, AWS

World Food Programme (WFP), India

- Built production RAG pipeline with **>85% retrieval accuracy** - reduced manual literature review effort by **~60%** for WFP research teams, enabling faster evidence synthesis and policy reporting.
- Delivered **3+ AI/ML PoCs** and 1 production document intelligence system for automated research extraction, structured analysis, and knowledge retrieval at scale

Ministry of Agriculture & Farmers Welfare (MoA&FW)

- Achieved **96% F1-score** on **100K+** agricultural images (EfficientNet, GCP) with a full MLOps pipeline - CI/CD, Docker, monitoring, and containerized deployment.
- Cut query response time by **~85%** via optimized RAG conversational AI; achieved **~92%** retrieval relevance on large unstructured agricultural document repositories.
- Delivered **5+ PoCs** and **2 production AI systems** - an advisory chatbot and a binary image classifier - for government-scale deployment on cloud infrastructure.

iNDX.Ai | Full Stack Developer *Jul'24 – Dec'24*

- Built **RESTful APIs** and full-stack apps (Node.js, PostgreSQL, Svelte) with RBAC, authentication, real-time interactions, and agile **CI/CD** workflows.

KEY PROJECTS

Twitter Sentiment Analysis - Deep Learning *Apr'23 – Jun'23*

- LSTM + GloVe classifier on 99K tweets - **78% accuracy** (vs 76.3% Naive Bayes baseline); robust preprocessing pipeline reduced noise by 35% and improved model convergence speed by 40%.
- Conducted ablation study comparing LSTM, BiLSTM, and CNN architectures across Word2Vec, GloVe, and FastText embeddings. Tech: Python, TensorFlow, LSTM, NLTK, FastAPI, Pandas

NIFTY-50 Stock Price Prediction - LSTM *Jan'23 – May'23*

- Achieved **R^2=0.84** with LSTM (tuned: LR, dropout, layers) and walk-forward validation; feature engineering with RSI, MACD, and Bollinger Bands. Tech: TensorFlow, Keras, Scikit-learn, yFinance

SKILLS & EXPERTISE

LLM & Gen AI	LangChain, LangGraph, RAG, Prompt Engineering, Agentic AI, OpenAI API, LLaMA, HuggingFace, FAISS, Vector DBs
ML / DL / CV	PyTorch, TensorFlow, Scikit-learn, EfficientNet, XGBoost, LSTM, CNN, OpenCV, Albumentations, Transfer Learning
MLOps & Cloud	Docker, CI/CD, GCP, AWS, Model Deployment, Streamlit, FastAPI, Git, Linux
Data & Analytics	Pandas, NumPy, SQL, EDA, Feature Engineering, Statistical Analysis, Matplotlib, Seaborn

LEADERSHIP & ACHIEVEMENTS

Associate Placement Manager | IIT Bombay Placement Team *Sep'22 – Dec'22*

- Led 300+ member team managing placements for 4,100+ students; achieved **14%** higher Day-1 selections via redesigned interview allocation model, saving 150+ man-hours; awarded **Certificate of Excellence**.
- Gold Medal in Carrom, IIT Bombay (2022) | Interests: LLMs, Generative AI, RAG, Agentic AI, Computer Vision, MLOps, Transformers